

What Is Secure Calendar?

A lightweight, encrypted scheduling application purpose-built for environments handling **Controlled Unclassified Information (CUI)**. Three independent security layers protect data at rest, in memory, and through secure erasure—all in a single desktop process with **zero network dependencies**.

Three-Layer Security

 **Layer 1: Encryption at Rest** AES-256 via SQLCipher
Argon2id KDF (64MB memory-hard) → 256-bit key → encrypted DB pages

 **Layer 2: Memory Protection** mlock'd Secure-String Buffers
OS cannot swap decrypted data to disk; WeakSet tracks all active buffers

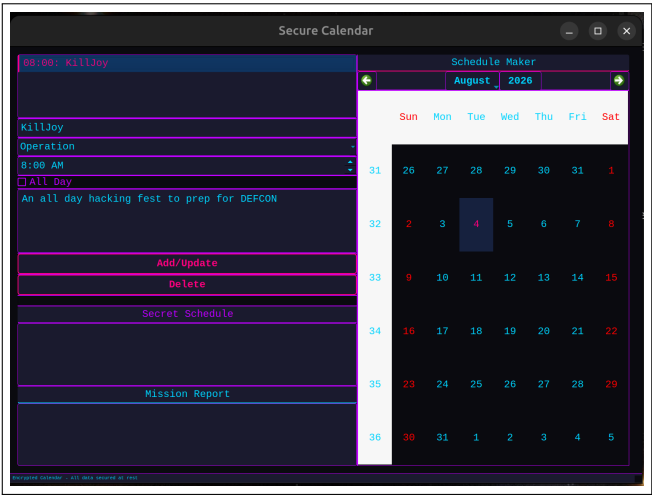
 **Layer 3: Secure Erasure** 8-Pass Wipe (3 RAND + 5 Zero)
OPENSSL_cleanse barrier; triggered on navigate, delete, clear, & shutdown

NIST Standards Alignment

Standard	Implementation
SP 800-88	8-pass secure wipe with compiler barrier
SP 800-132	Argon2id password-based key derivation
SP 800-131A	AES-256 approved encryption algorithm
SP 800-171	CUI media protection controls
SP 800-63B	Rate-limited passphrase authentication

Coded to NIST standards. Not NIST certified or approved.

Application Screenshot

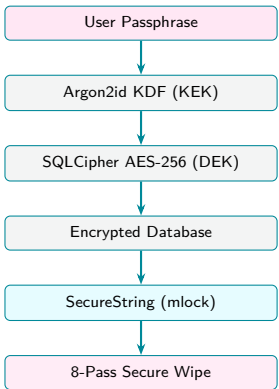


Status bar: "Encrypted Calendar – All data secured at rest"

Low SWaP Footprint

Metric	Value
Dependencies	2 pip packages
Disk	<15 MB (app code)
RAM	<80 MB runtime
Network	None (air-gap ready)
External Services	None required
Startup	<2 seconds

Key Architecture



Why Oblivion Edge?

- ✓ Defense-in-depth by default
- ✓ Deployable on air-gapped networks
- ✓ OpenSSL 3.5 with post-quantum readiness
- ✓ Scalable architecture patterns
- ✓ Coded to NIST standards

